Saion K. Roy

Postdoctoral Researcher

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Research Profile

My research develops secure and energy-efficient machine-learning systems through algorithm–architecture–circuit co-design. I work with in-memory computing and emerging accelerator architectures, emphasizing how algorithm mapping, microarchitecture, and silicon implementation shape efficiency, robustness, and the hardware attack surface. My goal is an end-to-end design methodology that treats security and privacy as first-class constraints alongside accuracy, energy, and reliability.

Research Areas Integrated Circuits, Machine Learning Hardware, Hardware Security

Education

Aug 2018 – Aug 2024
PhD, Electrical and Computer Engineering, *University of Illinois at Urbana-Champaign (UIUC)*, GPA: 3.88/4.0
Thesis: Energy-efficient Resistive In-memory Computing Architectures
Advisor: Professor Naresh R. Shanbhag
Committee: Pavan K. Hanumolu, Saugata Ghose, and Jae-sun Seo

July 2013 – July 2018
Bachelor and Master of Technology, Electronics and Electrical Communication Engineering
Indian Institute of Technology (IIT) Kharagpur, GPA: 9.26/10

Academic Appointments

Sep 2024 – present
Postdoctoral Research Associate, *Northeastern University*, Electrical and Computer Engineering
Advisor: Professor Yunsi Fei

Publications

2026 *SLAC: Access-Driven CPU-to-GPU Side-channel Attacks via System-Level Cache on Apple Silicon*, ACM SIGSAC Conference on Computer and Communications Security (CCS), 2026, to appear
Tianhong Xu[†], Saion K. Roy, Ruyi Ding, A. Adam Ding, and Yunsi Fei

2026 *ROBBIN: Rowhammer-Based Backdoor Injection during Inference*, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2026, to appear
Saion K. Roy[*], Yufei Wang[*†], A. Adam Ding, and Yunsi Fei

2026 *Attack from Shadows: Unsupervised Side-channel Transfer Learning across Devices and Modalities*, 21st ACM Asia Conference on Computer and Communications Security (AsiaCCS), 2026
Saion K. Roy, Ziyue Zhang, A. Adam Ding, and Yunsi Fei

2026 *HOTSPOT: Hardware Observation via Temperature Sensing for Protection against On-Chip Trojans*, IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I), 2026
Saion K. Roy, Thomas Gourousis[†], Mengting Yan, Junyi Yang, Ruyi Ding, Francesco Restuccia, Aatmesh Shrivastava, Marvin Onabajo, and Yunsi Fei

2026 *Comprehending the Energy-Accuracy-Security Trade-offs in Embedded Non-Volatile In-Memory Computing Architectures*, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2026
Saion K. Roy and Naresh R. Shanbhag

2024 *On the Security Vulnerabilities of MRAM-based In-Memory Computing Architectures against Model Extraction Attacks*, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2024
Saion K. Roy and Naresh R. Shanbhag

2024 *The Energy-Accuracy-Security Trade-off in Resistive In-memory Architectures*, IEEE International Electron Devices Meeting (IEDM), 2024
Saion K. Roy and Naresh R. Shanbhag

2024 — *Compute SNDR-boosted 22nm MRAM-based In-memory Computing Macro using Statistical Error Compensation*, IEEE Journal of Solid-State Circuits (JSSC), 2024
Saion K. Roy, Han-Mo Ou[†], Mostafa G. Ahmed, Peter Deaville, Bonan Zhang, Naveen Verma, Pavan K. Hanumolu, and Naresh R. Shanbhag

2024 — *Energy-Accuracy Trade-offs for Resistive In-Memory Computing Architectures*, IEEE Journal on Exploratory Solid-State Computational Devices and Circuits (JxCDC), 2024
Saion K. Roy and Naresh R. Shanbhag

2023 — *Compute SNR-boosted 22nm MRAM-based In-memory Computing Macro using Statistical Error Compensation*, IEEE European Solid-State Circuits Conference (ESSCIRC), 2023
Saion K. Roy, Han-Mo Ou[†], Mostafa G. Ahmed, Peter Deaville, Bonan Zhang, Naveen Verma, Pavan K. Hanumolu, and Naresh R. Shanbhag

2022 — *Benchmarking In-memory Computing Architectures*, IEEE Open Journal of the Solid-State Circuits Society (OJ-SSCS), 2022
Naresh R. Shanbhag and Saion K. Roy

2022 — *Comprehending In-memory Computing Trends via Proper Benchmarking*, IEEE Custom Integrated Circuits Conference (CICC), 2022
Naresh R. Shanbhag and Saion K. Roy

2022 — *Fundamental Limits on the Computational Accuracy of Resistive Crossbar-based In-memory Architectures*, IEEE International Symposium on Circuits and Systems (ISCAS), 2022 **Student Participation Grant Winner**
Saion K. Roy, Ameya D. Patil, and Naresh R. Shanbhag

[*] Equal contribution [†] Mentee

Manuscripts Under Submission

- Saion K. Roy, Kaining Zhou[†], Jian Huang, Yunsi Fei, and Naresh R. Shanbhag. *Title omitted to maintain anonymity*. In submission to the IEEE International Symposium on High-Performance Computer Architecture (**HPCA 2027**)
- Saion K. Roy[*], Kaining Zhou[*†], Soonha Hwang, Yunsi Fei, and Naresh R. Shanbhag. *Title omitted to maintain anonymity*. In submission
- Yufei Wang[†], Saion K. Roy, and Yunsi Fei. *Title omitted to maintain anonymity*. In submission to the Design, Automation and Test in Europe Conference (**DATE 2027**)
- Yashaswini Makaram, Davis Ranney, Amir Taherin, Saion K. Roy, David Kaeli, and Yunsi Fei. *Title omitted to maintain anonymity*. In submission to the Design, Automation and Test in Europe Conference (**DATE 2027**)

Grants and Sponsored Research

2026 — **Grant under preparation**
- **NSF SaTC 2.0: Towards Secure Inference of Mixture-of-Experts Large Language Models**, three-year collaborative project (PI Yunsi Fei, PI Ruyi Ding, and Co-PI Lili Su)
- Co-author and lead of the integrity thrust, which proposes causal selection of the experts that bridge an input trigger to the model output and selective model editing to backdoor MoE-based LLMs

Sep 2024 – present — **Sponsored projects contributed to as a postdoctoral researcher**
- **NSF Phase I IUCRC Northeastern University: Center for Hardware and Embedded System Security and Trust (CHEST)** (PI Yunsi Fei). Developed the cross-device unsupervised side-channel attack (AsiaCCS'26)
- **NSF RINGS: Internet of Things Resilience through Spectrum-Agile Circuits, Learning-Based Communications and Thermal Hardware Security** (PI Yunsi Fei). Demonstrated the on-chip temperature-sensor hardware-Trojan detector on a 65 nm ASIC (TCAS-I'26)
- **SRC/DARPA JUMP 2.0 Center for the Co-Design of Cognitive Systems (COCOSYS)** (PI Naresh R. Shanbhag). Analyzed the energy–accuracy–security trade-offs of resistive in-memory computing architectures, leading to papers in ICCAD'24, IEDM'24, and TCAD'26

Aug 2018 – Aug 2024 — **Sponsored projects contributed to as a graduate student**
- **DARPA ERI Foundations Required for Novel Compute (FRANC)** (PI Naresh R. Shanbhag). Built the resistive IMC behavioral models (JxCDC'24) and taped out the accuracy-boosted 22 nm MRAM IMC macro (ESSCIRC'23, JSSC'24)
- **SRC JUMP Center for Brain-inspired Computing (C-BRIC)** and **SRC/DARPA JUMP 2.0 COCOSYS** (PI Naresh R. Shanbhag). Developed the IMC benchmarking framework (CICC'22, OJ-SSCS'22) and derived the compute-SNDR limits of resistive crossbars (ISCAS'22)

Fellowships **DAAD WISE Fellowship**, German Academic Exchange Service, 2017
IEEE ISCAS'22 Student Participation Grant

Invited Talks and Presentations

2026
CAPER: Circuit-Aware Parameter-Efficient Reasoning in LLMs for Analog Verification
- *IBM/IEEE AI Compute Symposium (AICS'26)*, IBM T.J. Watson Research Center, Yorktown Heights, New York, USA (2026)

2025
Energy-Accuracy-Security Trade-offs in Resistive In-Memory Computing Architectures
- *New England Hardware Security Day*, MIT, Cambridge, Massachusetts, USA (2025)
- *SRC JUMP 2.0 Center for the Co-Design of Cognitive Systems Annual Review*, Georgia Tech, Atlanta, Georgia, USA (2025)
- *RAAINS Workshop at MIT Lincoln Laboratory (MITLL)*, Lexington, Massachusetts, USA (2025)

2024 –2025
Energy-Accuracy Trade-offs for Resistive In-Memory Computing Architectures
- *JxCDC SSCS Open Journal Webinar*, online (2025)
- *Indian Institute of Technology (IIT) Madras*, Chennai, India (2024)
- *Center for the Co-Design of Cognitive Systems*, online (2024)

2024
Security Vulnerabilities of eNVM-based In-Memory Architectures
- *Center for the Co-Design of Cognitive Systems*, online (2024)

2024
Compute SNDR-boosted 22nm MRAM IMC using Statistical Error Compensation
- *Center for the Co-Design of Cognitive Systems*, online (2024)

2024
GPLeak: Software Power Side Channel Leakage in GPUs 🏆 **Oral Presentation Award**
- *CSL Student Conference, Security and Privacy Session*, UIUC, Urbana, Illinois, USA (2024)

2023
Fundamental Limits on the Compute SNR of Resistive In-Memory Architectures 🏆 **Best Speaker Award**
- *CSL Student Conference, Circuits and Systems Session*, UIUC, Urbana, Illinois, USA (2023)

Research Experience

Featured
COCOSYS spotlight
NanoSI profile
NanoSI video

My research connects machine-learning security with integrated-circuit and system design across algorithms, computer architecture, and hardware implementation. Improving the energy efficiency of ML accelerators, for example through IMCs, exposes new attack surfaces. Commercial platforms show the same pattern, where shared resources such as caches and DRAM enable new side-channel and backdoor attacks. As these vulnerabilities cut across the stack, finding and fixing them calls for algorithm–architecture–circuit co-design, toward secure and efficient ML hardware.

2026
Collaborators
A. Adam Ding
Yunsi Fei
AsiaCCS 2026,
& CCS 2026

Side-channel attacks on consumer hardware Security
- Built a CPU-to-GPU Prime+Probe side channel on the Apple M1 shared system-level cache by reverse-engineering its indexing function
- Recovered over 90% of GNN edges and LLM input keywords with up to 94.8% accuracy from an unprivileged CPU process
- Proposed an unsupervised transfer-learning-based side-channel attack (UTLA) that recovers cryptographic keys across FPGA, microcontroller, and x86 targets, using as few as 61 traces

2026
Collaborators
A. Adam Ding
Yunsi Fei
ICCAD 2026
& Ongoing work

Backdoor attacks on ML accelerators Security
- Activated DNN backdoors through Rowhammer bit flips in DDR4 DRAM, reaching nearly 90% attack success across three chips, including CPU-to-iGPU attacks on shared-DRAM heterogeneous SoCs
- Showed that GPU-family numerical behavior alone can trigger backdoors in diffusion transformers, and investigated chip-fingerprint triggers in 28 nm digital IMC accelerators

2026
Collaborators
A. Adam Ding
Aatmesh Shrivastava
Francesco Restuccia
Marvin Onabajo
Yunsi Fei
ICCAD 2026
TCAS-I 2026,
& Ongoing work

Defenses against integrity and confidentiality attacks Security
- Proposed vulnerability-aware DRAM page allocation that keeps model weights off Rowhammer-prone pages, removing all bit flips and restoring clean accuracy from 86% to 91%
- Proposed a compiler–GPU co-design that fixes the order and rounding of reductions across GPUs, preventing trigger generation and platform-specific backdoors
- Detected hardware Trojans in a 65 nm ASIC by repurposing existing on-chip differential temperature sensors, reaching up to 95% accuracy with $6.65\times$ higher SNR than supply-current side-channel monitoring

2024–2026
Collaborators
Naresh R. Shanbhag
ICCAD 2024,
IEDM 2024,
& TCAD 2026

Energy–accuracy–security trade-offs in resistive IMCs Circuits Security
- Showed that analog noise lowers compute SNDR but does not protect stored weights from gradient-based model extraction attacks (MEAs)
- Recovered weights from the 22 nm MRAM prototype that achieve ResNet-20 accuracy within 0.1% of baseline at a bank-level SNDR of 4.18 dB
- Identified a secure operating zone below 2 dB compute SNDR, robust to MEAs, where noise-aware training or SEC can restore accuracy for the legitimate user

2023–2024

Accuracy-boosted resistive IMC Circuits

Collaborators
Naveen Verma
Pavan K. Hanumolu
Naresh R. Shanbhag
ESSCIRC 2023,
& JSSC 2024

- Designed a 22 nm MRAM IMC macro with offset-compensating current sensing for mismatch and statistical error compensation (SEC) for wire parasitics
- Measured a 2.7 dB–6 dB compute-SNDR gain at 0.8% energy and 12.2% area overhead
- Increased ResNet-20 accuracy on CIFAR-10 from 74.8% to 82.0% without noise-aware training

2022–2024

Resistive IMC modeling Circuits

Collaborators
Naresh R. Shanbhag
ISCAS 2022,
& JxCDC 2024

- Developed statistical signal-and-noise models of crossbar and parallel-bar IMCs for MRAM, ReRAM, and FeFET, validated in a commercial 22 nm process
- Showed that raising the device conductance contrast beyond about 12 gives diminishing returns in compute accuracy for crossbar and parallel-bar IMCs, as device non-idealities and sensing noise begin to dominate
- Compute-SNDR maxes out at 18 dB–22 dB in parallel-bars, set by parasitics, conductance variation, and sensing mismatch, which also maximizes the network-level accuracy

2022

IMC benchmarking Circuits

Collaborators
Naresh R. Shanbhag
CICC 2022,
& OJ-SSCS 2022

- Built a compositional framework that compares over 100 IMC and digital chips using bit-normalized TOPS, TOPS/W, and TOPS/mm^2, released as an open resource
- Showed that SRAM IMCs lead digital accelerators by $13\times$ in bank-level energy efficiency and $6.6\times$ in compute density, while resistive IMCs lag in both

Industry and Research Internships

2021

Research Intern, *Texas Instruments, Kilby Labs*, Dallas, USA, Summer
Supervised by: Dr. Mahesh Mehendale and Dr. Avishek Biswas

- **NAND Flash-based IMC Design**: Characterized NAND flash cell properties and designed sensing circuits for IMC operation
- **Behavioral Modeling of ADC in PyTorch**: Analyzed the impact of ADC non-idealities on the DNN accuracy of IMC implementations using PyTorch models

2017

Research Intern 🏆 **DAAD WISE Fellow**, *Technische Universität Berlin (TU Berlin)*, Berlin, Germany, Summer
Supervised by: Prof. Dr.-Ing. Friedel Gerfers

- **SAR-ADC Digital Block Optimization**: Reduced power consumption of a 10-bit, 100 MS/s SAR ADC digital block from 1.7 mW to 1.1 mW in 65 nm TSMC technology through circuit- and logic-level optimizations
- **Low-Power Flip-Flop Study**: Evaluated flip-flop architectures for power-delay trade-offs. Adopted transmission gate-based pulsed flip-flops achieving 50% power reduction over conventional designs

2016

Research Intern, *Semiconductor Laboratory, Dept. of Space, Govt. of India*, Chandigarh, India, Summer
Supervised by: H. S. Jattana

- **Two-Stage Op-Amp Design**: Designed and characterized a two-stage op-amp (73 dB gain, 50 MHz bandwidth, 55° phase margin, 20 V/μs slew rate) in 180 nm CMOS
- **Clocked Comparator Design**: Designed a latch-based clocked comparator (50.37 dB gain, 4.7 ns delay) at 100 MHz with 2.5 mW power consumption

Teaching

Fall 2025

EECE 5699: Computer Hardware and System Security, *Northeastern University*, Co-Course Instructor

- **Primary Instructor**: Prof. Yunsi Fei
- Delivered lectures and prepared homework for graduate students, connecting hardware vulnerabilities to their system-level consequences

Fall 2022

ECE 342: Electronic Circuits, *University of Illinois at Urbana-Champaign (UIUC)*, Teaching Assistant

- **Instructors**: Prof. Pavan K. Hanumolu and Prof. Chandrasekhar Radhakrishnan
- Conducted office hours, guiding students hands-on, walking them step by step through circuit analysis and design problems, and prepared homework and exams

Spring 2018

EC39004: VLSI Laboratory, *Indian Institute of Technology (IIT) Kharagpur*, Teaching Assistant

- Senior-year laboratory paired with the VLSI design coursework
- Guided students through setting up and interpreting device simulations in Silvaco, and assessed their laboratory work

Fall 2017

EC29003: Introduction to Electronics Lab, *Indian Institute of Technology (IIT) Kharagpur*, Teaching Assistant

- Large sophomore-year laboratory in basic electronics
- Mentored students as they built, measured, and debugged circuits at the testbench, and evaluated their laboratory work

Professional Service

Session Chair
- IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2026
- 21st ACM Asia Conference on Computer and Communications Security (AsiaCCS), 2026

Technical Program Committee
- IEEE/ACM International Symposium on Computer Architecture (ISCA), 2026
- IEEE/ACM International Symposium on Microarchitecture (MICRO), 2026
- IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2026

Journal Reviewer
- IEEE Journal of Solid-State Circuits (JSSC)
- IEEE Open Journal of Circuits and Systems (OJCAS)
- IEEE Transactions on Very Large Scale Integration (VLSI) Systems (TVLSI)
- IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)
- IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I)
- IEEE Transactions on Circuits and Systems II: Express Briefs (TCAS-II)
- IEEE Journal on Emerging and Selected Topics in Circuits and Systems (JETCAS)
- ACM Transactions on Architecture and Code Optimization (TACO)
- IEEE Transactions on Electron Devices (TED)
- IEEE Design & Test (D&T)

Conference (External) Reviewer
- IEEE International Symposium on Circuits and Systems (ISCAS)
- ACM SIGSAC Conference on Computer and Communications Security (CCS)
- ACM/IEEE Design Automation Conference (DAC)
- Network and Distributed System Security Symposium (NDSS)

Mentoring

- **Tianhong Xu**: PhD Student, Northeastern University
- **Yufei Wang**: PhD Student, Northeastern University
- **Thomas Gourousis**: PhD Student, Northeastern University
- **Yashaswini Makaram**: PhD Student, Northeastern University
- **Han-Mo Ou**: PhD Student, University of Illinois at Urbana-Champaign
- **Kaining Zhou**: PhD Student, University of Illinois at Urbana-Champaign

Technical Skills

Programming	Python (PyTorch), Verilog, Verilog-A, MATLAB, C
EDA Tools	Cadence Virtuoso, Genus, Innovus, SPICE, Xilinx Vivado
Testing Tools	KiCad (PCB design), PYNQ, Arduino, and Atmel AVR test boards

Professional References

Prof. Naresh R. Shanbhag
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University of Illinois at Urbana-Champaign
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Prof. Yunsi Fei
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